PAVAN DHARMA ADAPA

Research Data Analyst · Biomedical Informatics

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PROFESSIONAL SUMMARY

M.S. in Computer Science graduate focused on research data. Co-author on a review manuscript under revision. As a graduate research assistant, cleaned and analyzed a 900,000+ record livestock-disease outbreak dataset for Computer Science and Veterinary Medicine faculty. Contributed 36 merged pull requests to a lab's open-source DNA sequencing pipeline. Analyzes public CMS claims filings and CDC survey microdata in SQL and Python.

TECHNICAL SKILLS

Research Data Practice: Data management and procurement, survey-weighted estimation, validation against independent implementations, provenance and documentation, manuscript preparation

Analysis & Programming: Python (pandas, NumPy, SciPy, scikit-learn), R, SQL, Excel, statistical modeling

Scientific Computing: Oxford Nanopore 16S rRNA sequencing data, NCBI reference databases, Snakemake workflows, Git and continuous integration, Linux, macOS, WSL2

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University*, Starkville, MS — Jul 2026 – Present

- Contributed **36 merged pull requests** to the lab's open-source DNA sequencing pipeline (maintained by Li Zhang), making it a documented, tested tool that installs in one command and runs on Linux, macOS, WSL2 and Windows.
- Validated pipeline output against an independently built implementation, **agreeing to within 0.01 percentage points** on every species and genus.
- Processed and quality-controlled **2M+ sequencing reads across 19 samples**, and automated construction of the NCBI reference database into a single command.

Graduate Teaching Assistant | *Mississippi State University*, Starkville, MS — Sep 2025 – May 2026

- Taught data cleaning, transformation and visualization in Python and R to **100+ students across six courses**.

Graduate Research Assistant | *Mississippi State University*, Starkville, MS — Aug 2024 – Aug 2025

- Cleaned and analyzed a **900,000+ record** Vesicular Stomatitis Virus (VSV) livestock-outbreak dataset, and presented the findings to Computer Science and Veterinary Medicine faculty.
- Co-author (third of five) on a review manuscript under revision after peer review; contributed the data preparation, statistical modeling and results reporting, and drafted technical sections.
- Built a Python/Selenium scraper that collected genome metadata for a poultry respiratory-disease project, **cutting collection from hours to under 10 minutes per dataset**.

HEALTH DATA PROJECTS

Emergency Department Throughput — *CDC/NCHS national survey microdata*

- Analyzed **16,025 real ED visits** survey-weighted to 155,397,747 national visits; the pipeline refuses to run unless it reproduces CDC's published total, so the figures cannot drift undetected.
- Showed between-hospital variation in door-to-provider time is **3.56x larger** than between-hour variation, and reported the expected peak-hour effect as a null result rather than dropping it.

Clinical Quality Measures Engine — *HEDIS / CMS eCQM · synthetic non-PHI EHR (Synthea)*

- Computed HEDIS and CMS eCQM measures over a **13,810-patient EHR** (820,993 encounters, 10.8M observations), holding each measure as a versioned specification with its SNOMED CT and LOINC value sets separate from the engine — so an annual steward update is a reviewable diff a clinical analyst signs off, not a code change.
- Traced **203 of 204** diabetes-measure failures to missing HbA1c tests, an artifact of how the Synthea generator schedules labs, and reported it as an engine check rather than a finding about care.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS — May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India — May 2024

Publication (under revision): Li Zhang, Saikanth Ratnavale, Pavan Dharma Adapa, Sai Harika Gade, Michael E. Navicky — Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production.